

1. Create a package com.math.operations and write a class to perform addition and subtraction.

```
package com.math.operations;
class Calcu{
    int add(int a, int b) {
        return a+b;
    }
    int sub(int a, int b) {
        return a-b;
    }
}
public class Calculator {
    public static void main(String[] args) {
        // TODO Auto-generated method stub

        Calcu c=new Calcu();
        int sum = c.add(10, 5);
        int sub = c.sub(10, 5);
        System.out.println("Addition : " + sum);
        System.out.println("Subtraction : " + sub);

    }
}
```

```
Addition : 15
Subtraction : 5
```

2. Create an abstract class Bank with method getInterestRate() and implement it in different bank classes.

```
package com.math.operations;
abstract class Bank{
    abstract double getInterest();

    void display() {
        System.out.println("Bank Details");
    }
}
class SBI extends Bank{
    double getInterest() {
        return 5.5;
    }
}
class CBI extends Bank{
    double getInterest() {
        return 4.5;
    }
}
```

```

public class BankTest {
    public static void main(String[] args) {
        // TODO Auto-generated method stub

        Bank b1 = new SBI();
        Bank b2 = new CBI();

        b1.display();
        System.out.println("SBI Interest Rate: " + b1.getInterest() + "%");
        System.out.println("CBI Interest Rate: " + b2.getInterest() + "%");
    }
}

```

Bank Details

SBI Interest Rate: 5.5%

CBI Interest Rate: 4.5%

3. Write a program where an interface is implemented and exception handling is used inside the method.

```

package Exception_handling;
interface Payment{
    void transaction(double amt, int member);
}
class Bill implements Payment{
    public void transaction(double amt, int member) {
        try {
            double amount = amt/member;
            if (member == 0) {
                throw new ArithmeticException("Division by zero");
            }
            System.out.println("Each member pay: " + amount);
        }
        catch (ArithmeticException e){
            System.out.println("Catch block");
            System.out.println("Check code: " + e.getMessage());
        }
        catch (Exception e) {
            System.out.println("Exception" + e);
        }
    }
}

public class ExceptionPractice {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Payment myProcessor = new Bill();
        System.out.println("No exception---");
        myProcessor.transaction(1000.0, 5);
        System.out.println("Exception----");
        myProcessor.transaction(500.0, 0);
    }
}

```

```
    }  
}  
No exception---  
Each member pay: 200.0  
Exception----  
Catch block  
Check code: Division by zero
```